Effect of Internal Geometry on Strength in 3D-Printed Implants

Comparing simple cubic and body-centered cubic lattices

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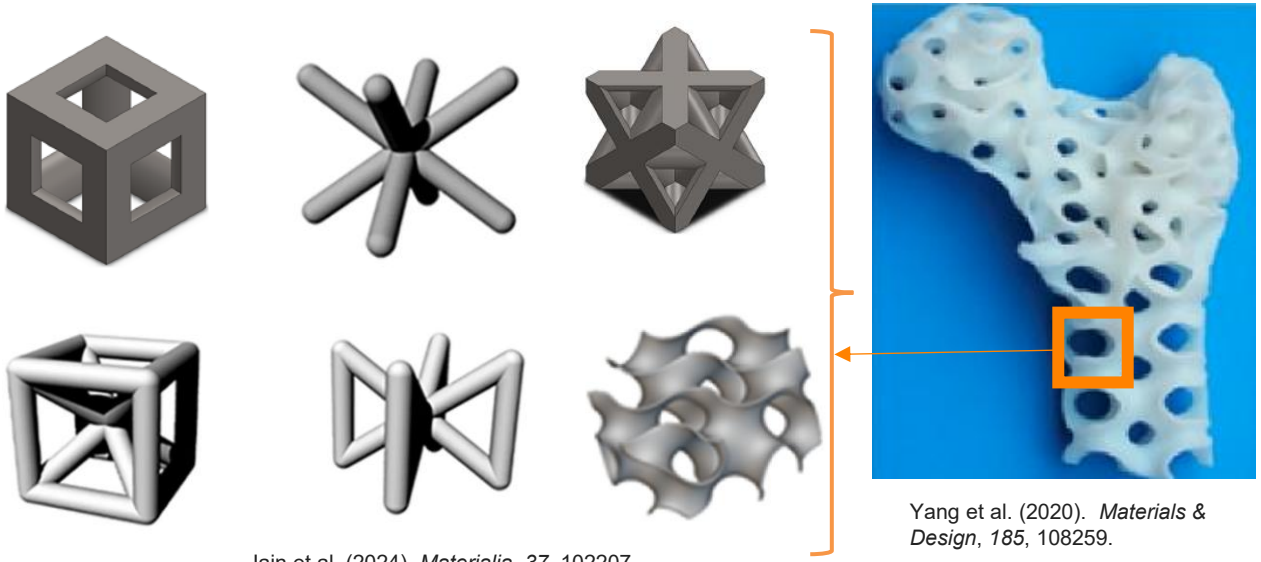
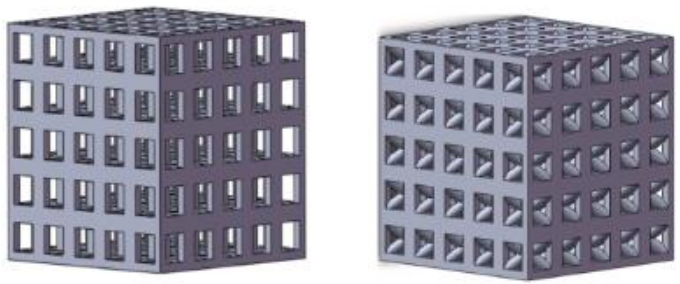
Objective

- Understand how internal geometry affects strength in 3D-printed scaffolds and implants
- Compare two common scaffold geometries:
  - Simple Cubic (SC)
  - Body-Centered Cubic (BCC)
- Use simulation instead of trial and error
- Study how they deform under compression

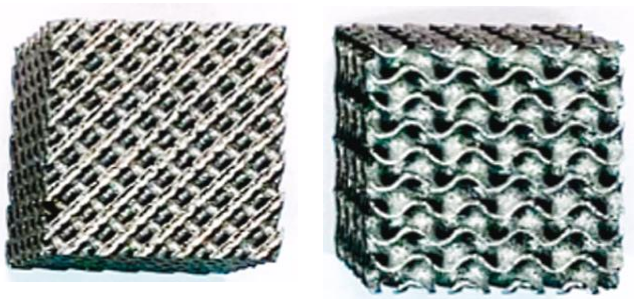


Jain et al. (2024). *Materialia*, *37*, 102207.
Chen et al. (2020). *Frontiers in bioengineering and biotechnology*, *8*, 609.

Yang et al. (2020). *Materials & Design*, *185*, 108259.



Zhen et al. (2025). *Polymer*, *333*, 128665.

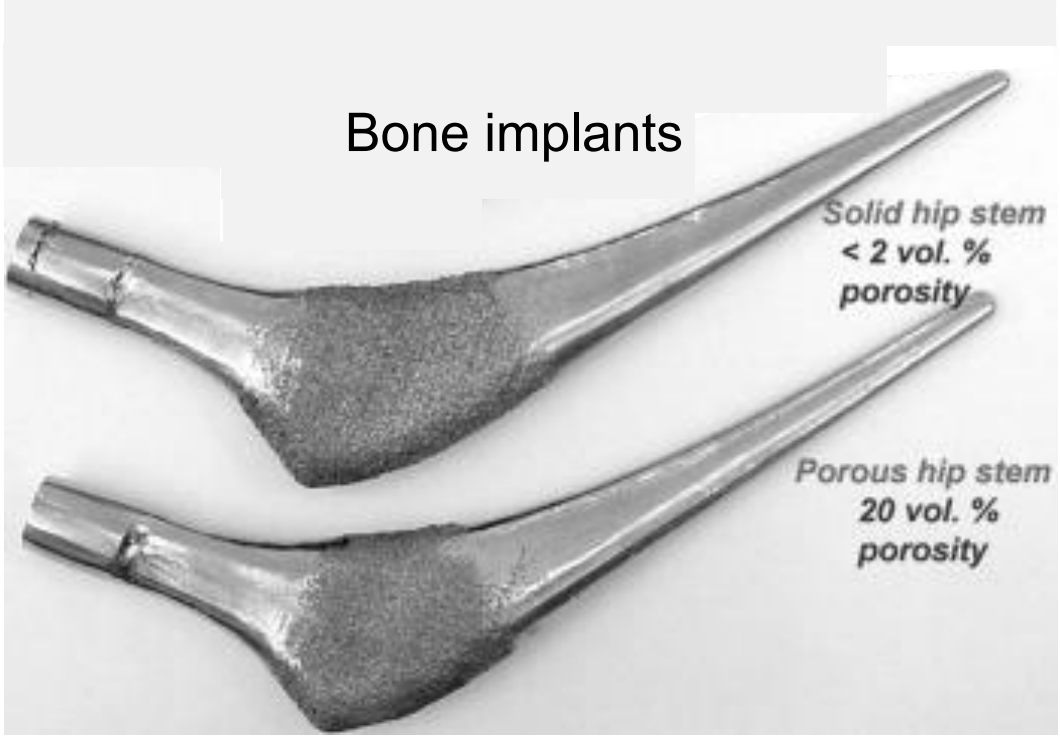


Seehanam et al. (2024). *Heliyon*, *10*(4).

Current Approach & Limitations

- Traditional implants are too stiff
- This causes stress shielding and poor bone integration
- Porous lattice structures can improve performance
- Current design relies on trial and error
- Limited understanding of geometry effects
- Physical testing is expensive

Internal geometry directly controls implant performance



Krishna et al. (2007). *Acta biomaterialia*, *3*(6), 997-1006.

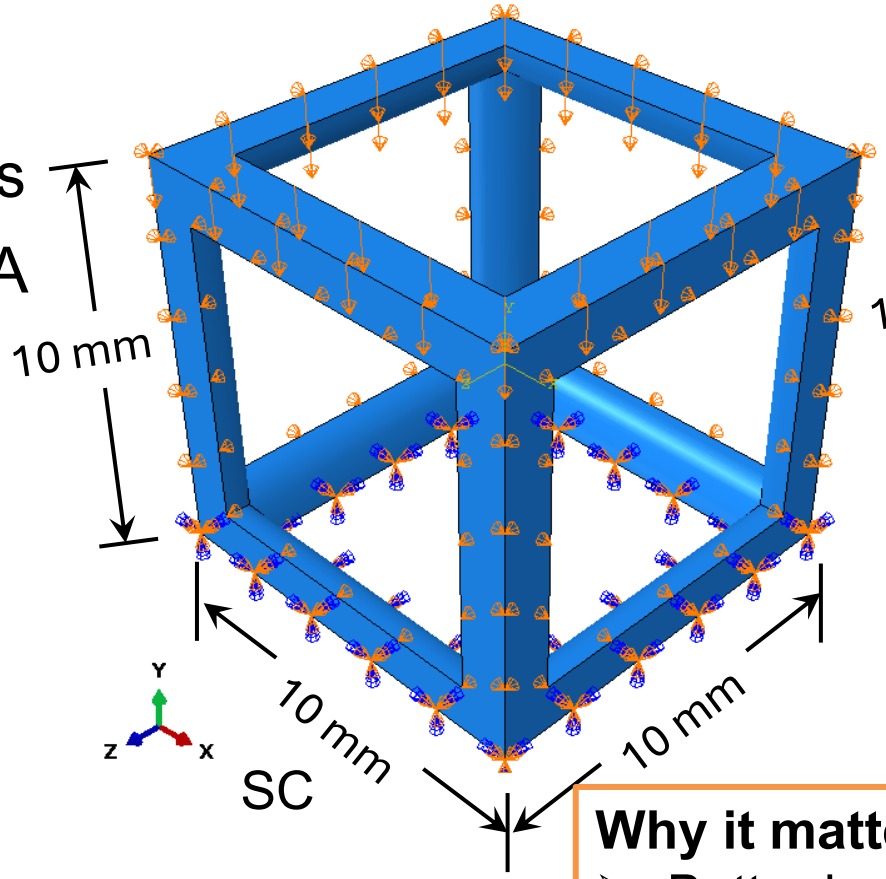
Porosity improves load transfer

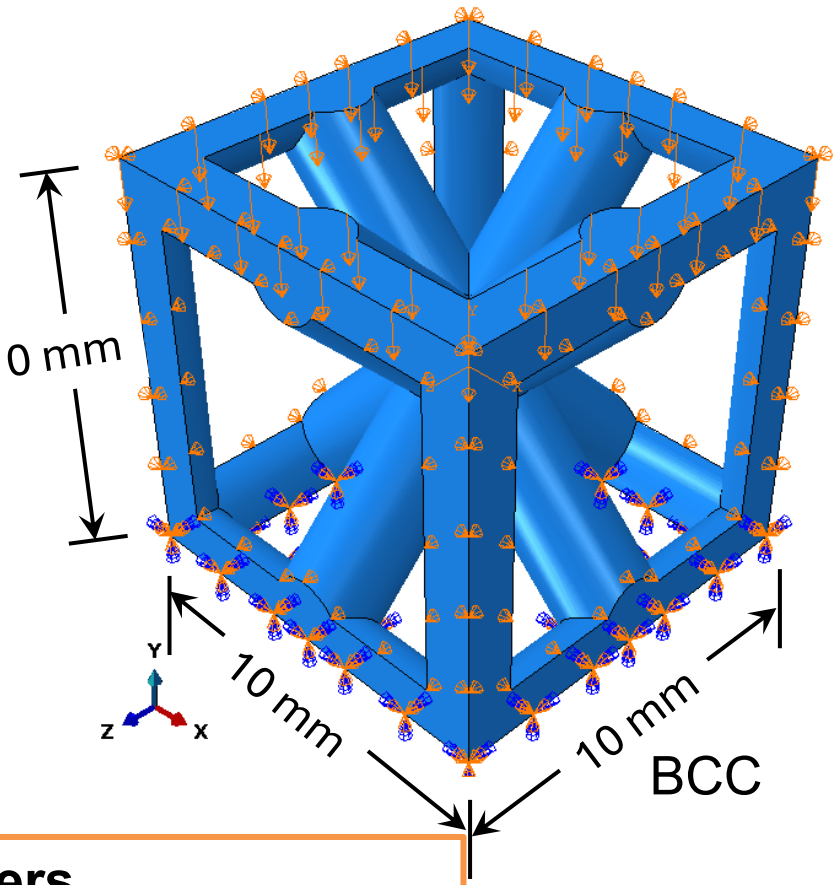
Approach

- Only Geometry changes
- Same material (PLA from literature):

Senatov et al. (2016). *Journal of the mechanical behavior of biomedical materials*, *57*, 139-148.

  - E = 1439 Mpa
  - $\upsilon = 0.3$
  - Yield stress: 21 MPa
  - Elastic-plastic
  - Isotropic hardening
- Same loading





Isolates the effect of internal geometry on mechanical response

Why it matters

- Better implant performance
- Reduced stress shielding
- Faster design process
- Potential cost reduction

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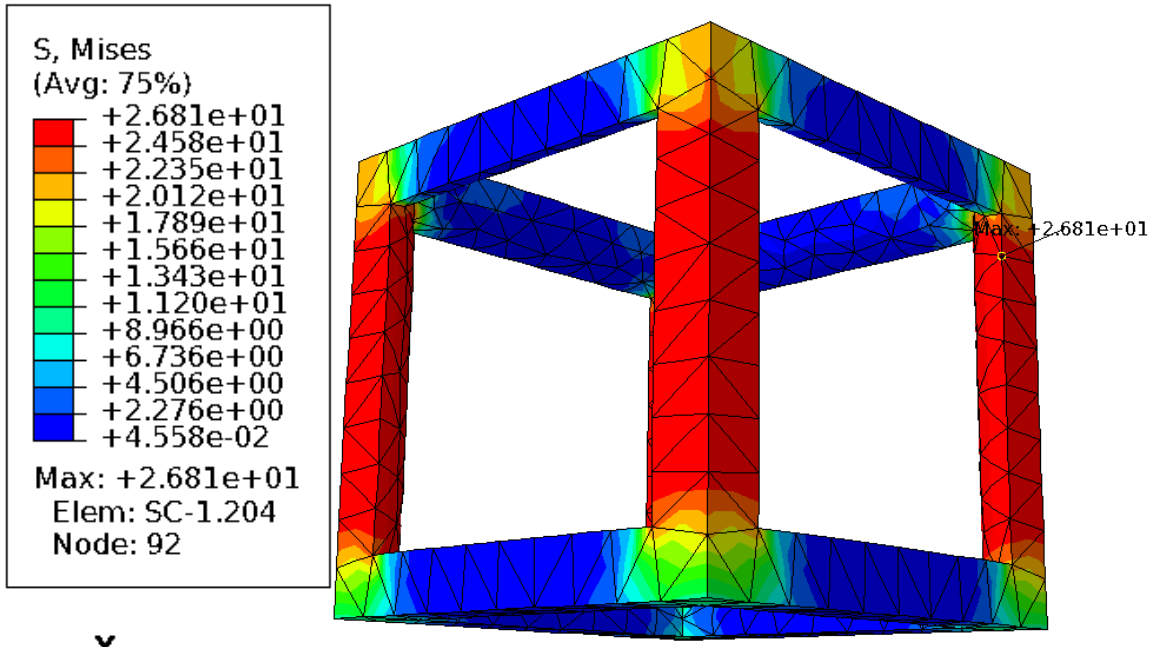
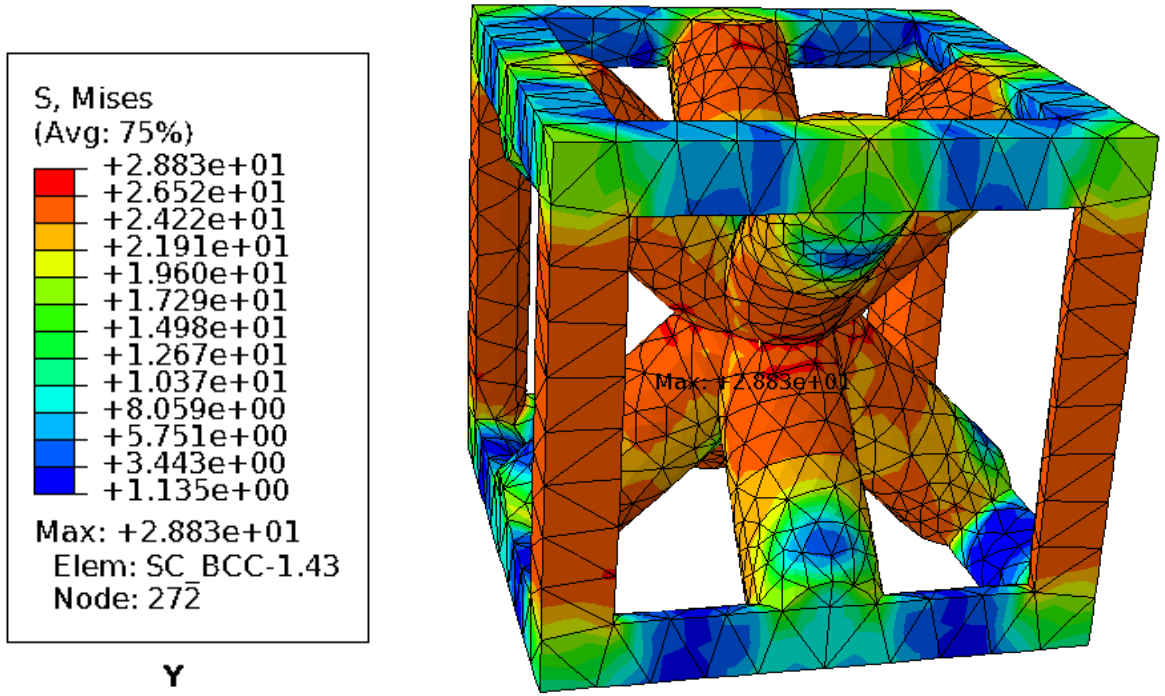
Effect of Geometry (Single Cell)



Lower stiffness, stress concentration at vertical members



Higher stiffness, load distributed through multiple load paths

- Geometry changes the location and magnitude of local stress concentrations.
- Geometry controls stiffness through load path connectivity.

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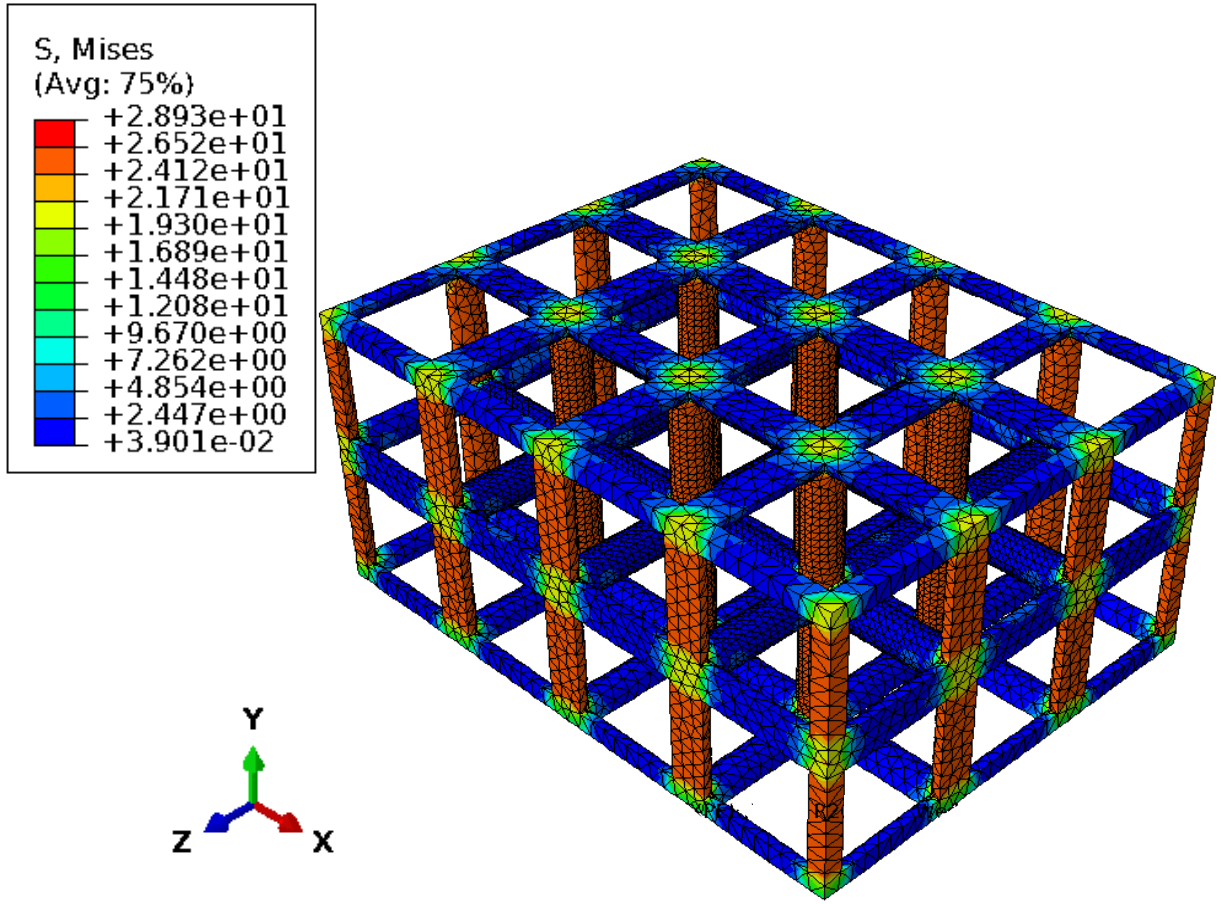
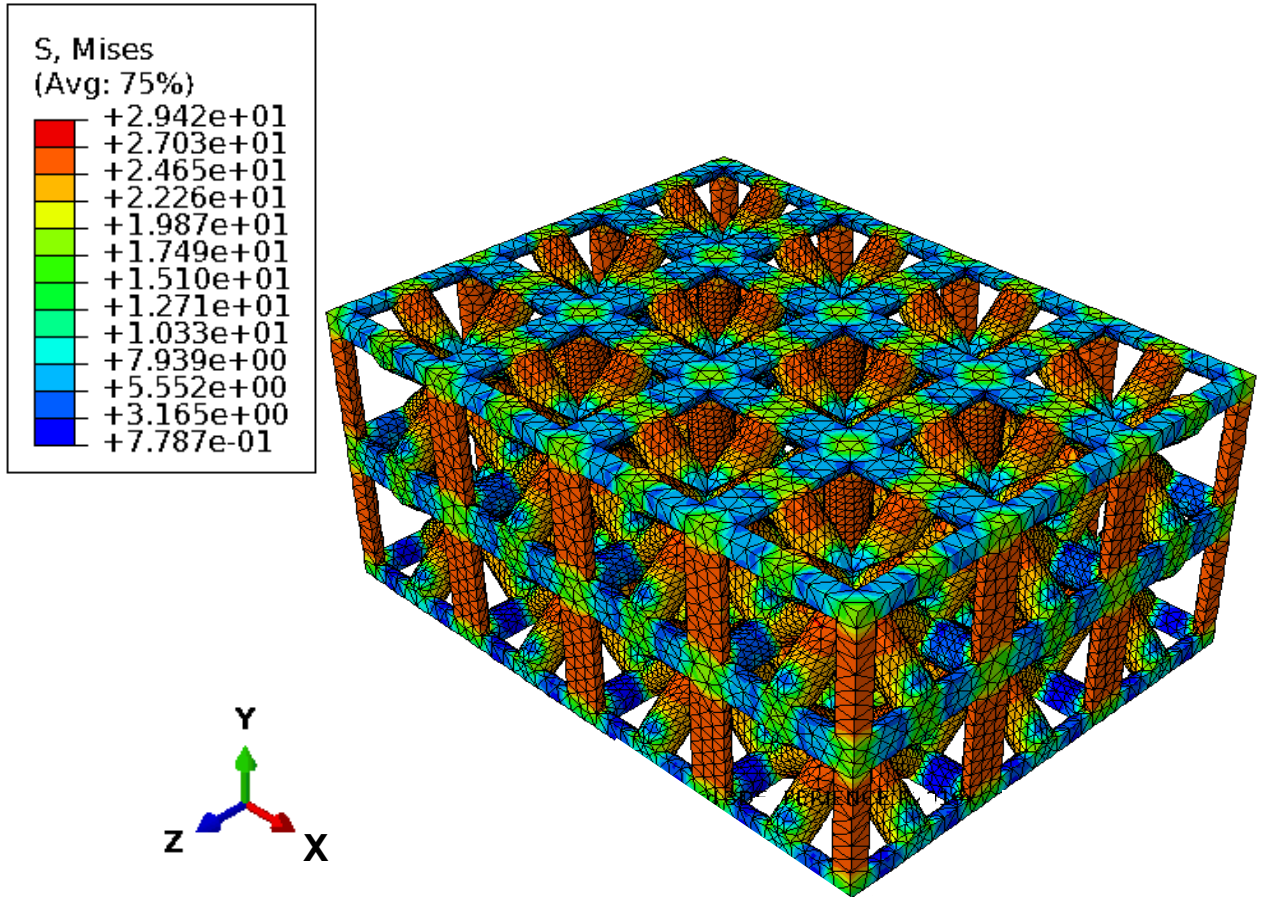
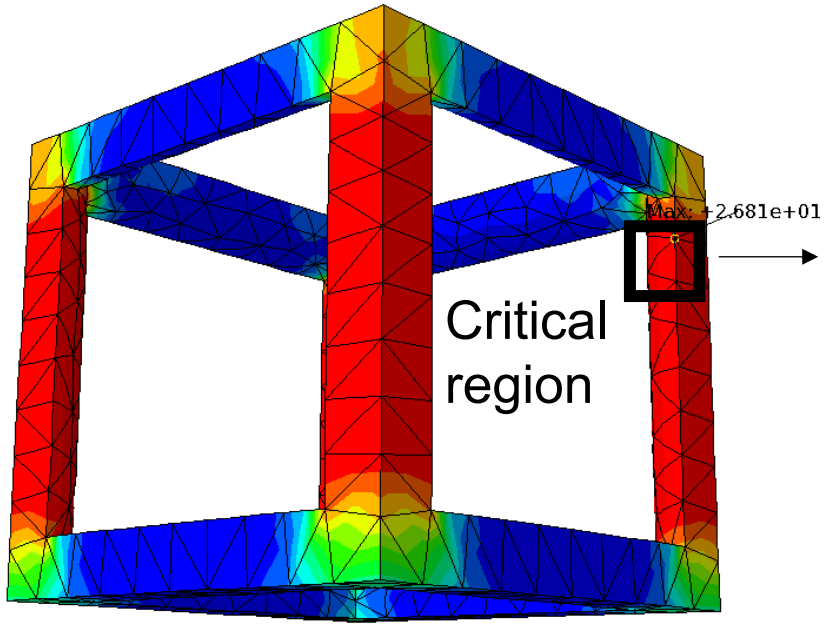
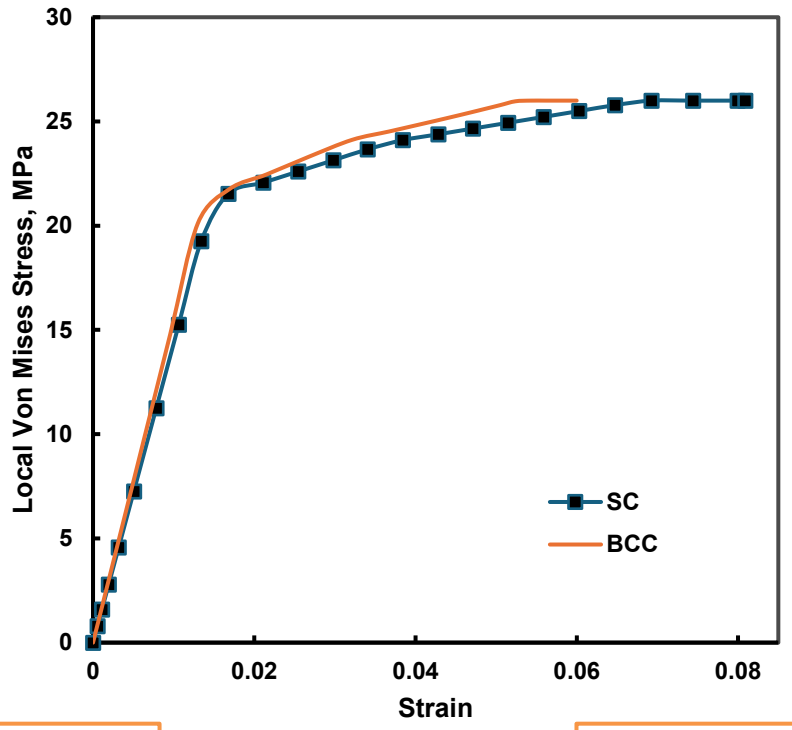
Effect of Geometry (Lattice Scale)

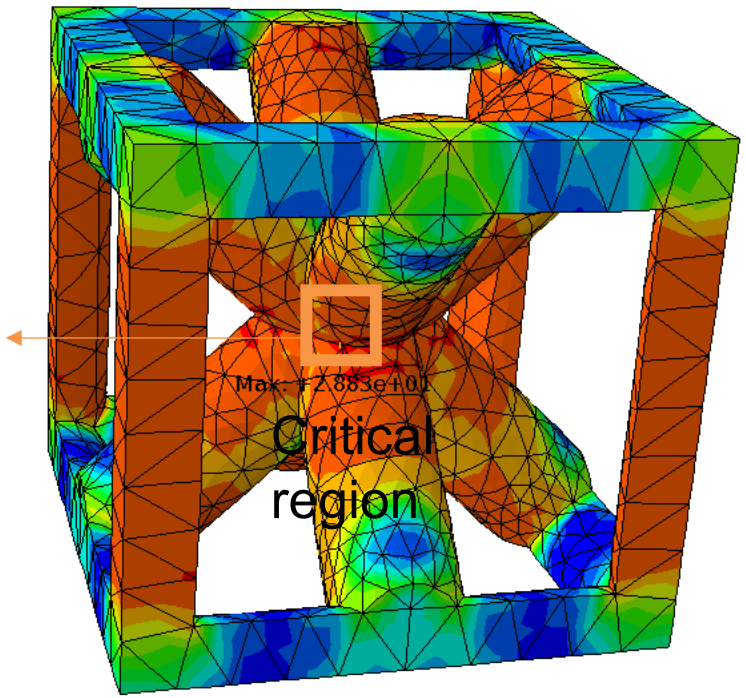




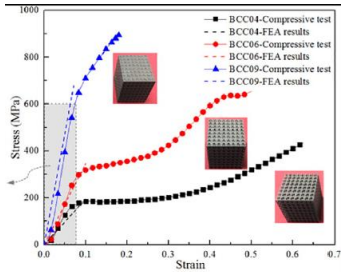
Validation & Risk







Qualitative Validation:
The local stress-strain curve captures elastic plastic response that is available in Literature. Wang et al. (2017). *Materials & Design*, *133*, 62-68.



Risks:
- Simplified material model
- Ideal geometry (no manufacturing defects)

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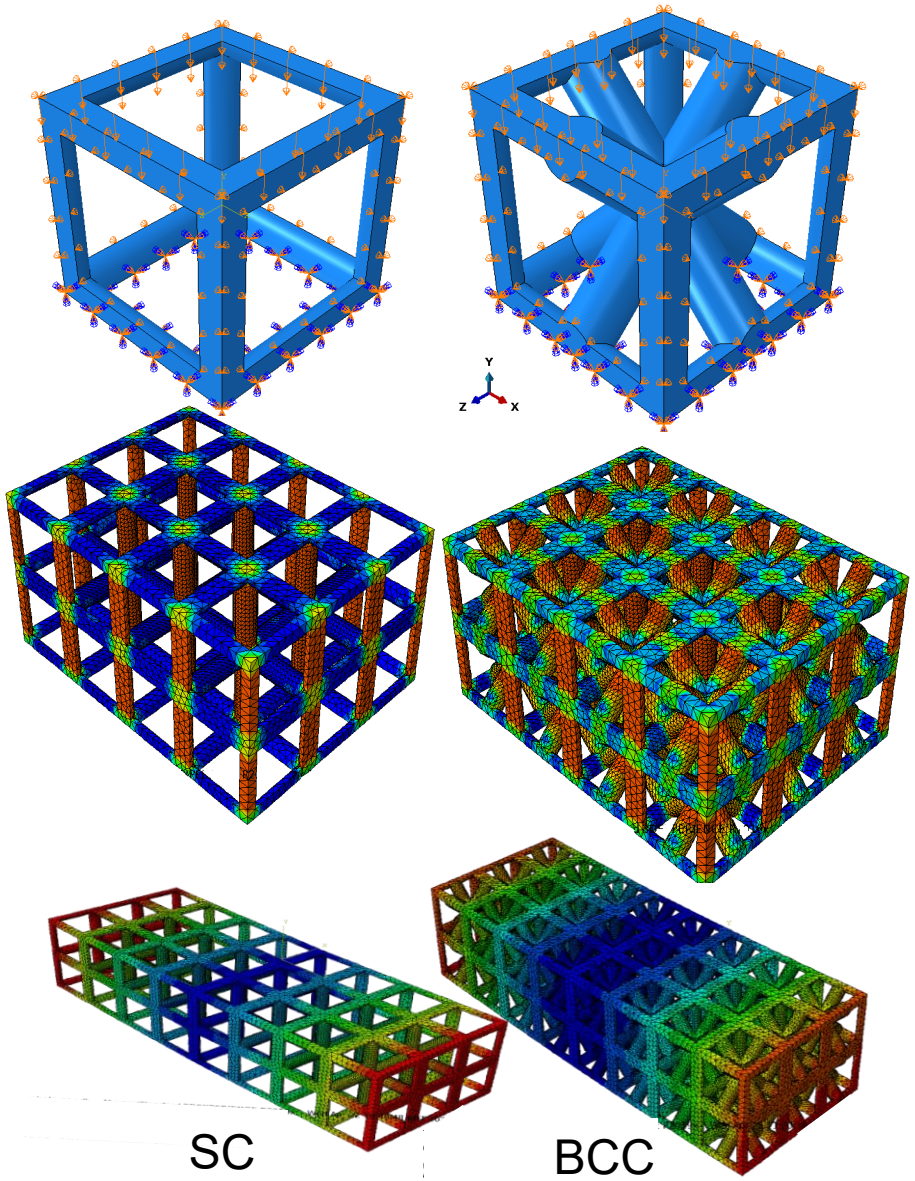
Conclusion

- Geometry significantly controls mechanical response
- SC:
  - Localized stress
  - Fewer load paths
- BCC:
  - Distributed load
  - Higher Stiffness
- Simulation enables faster and more reliable design



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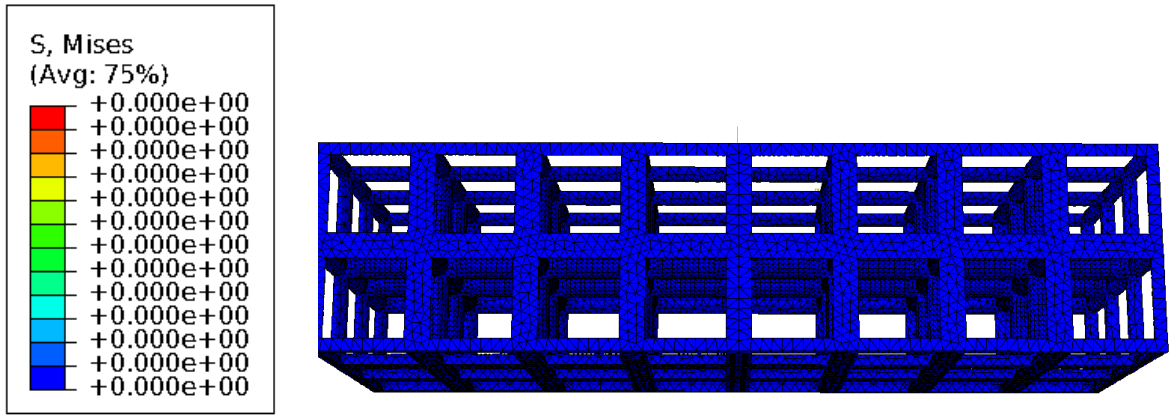
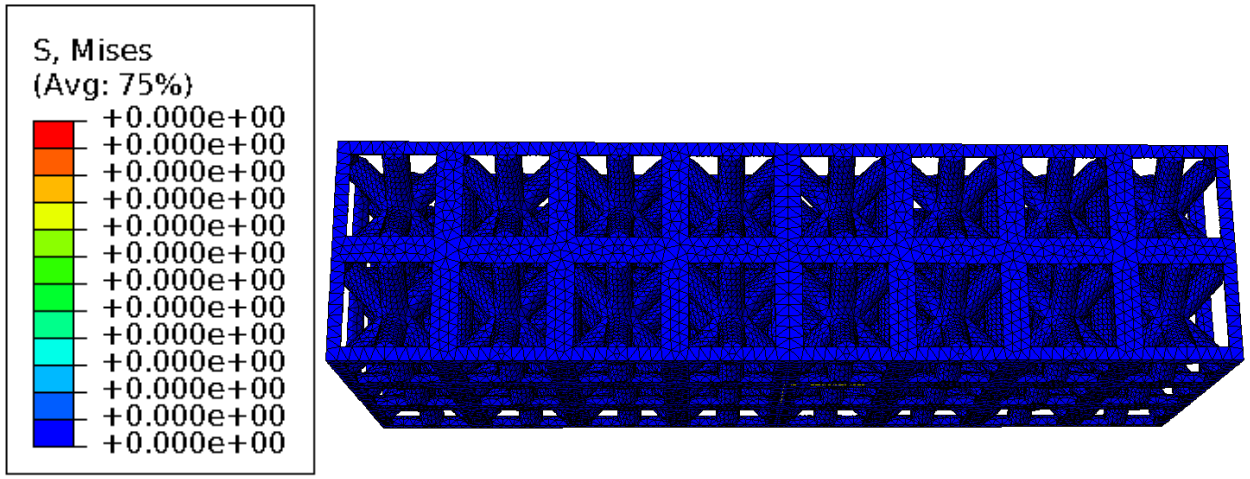
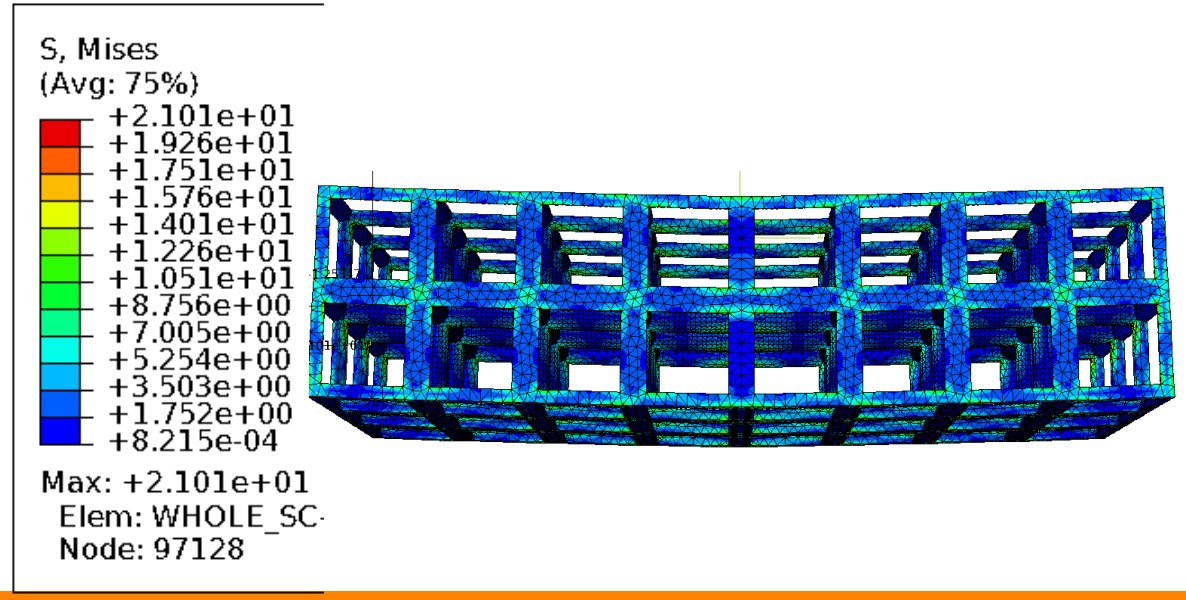
Thank you

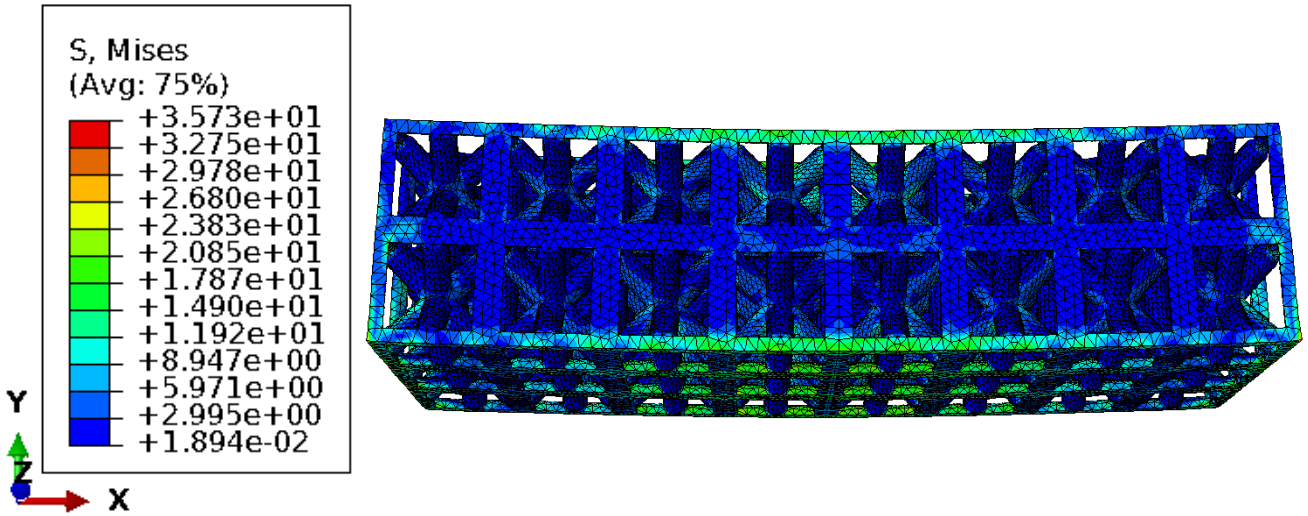
Questions?

Simplified Three-point Bending









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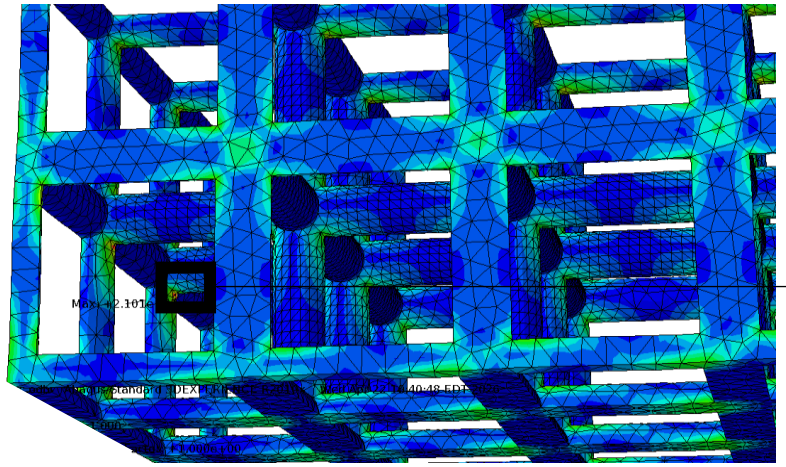

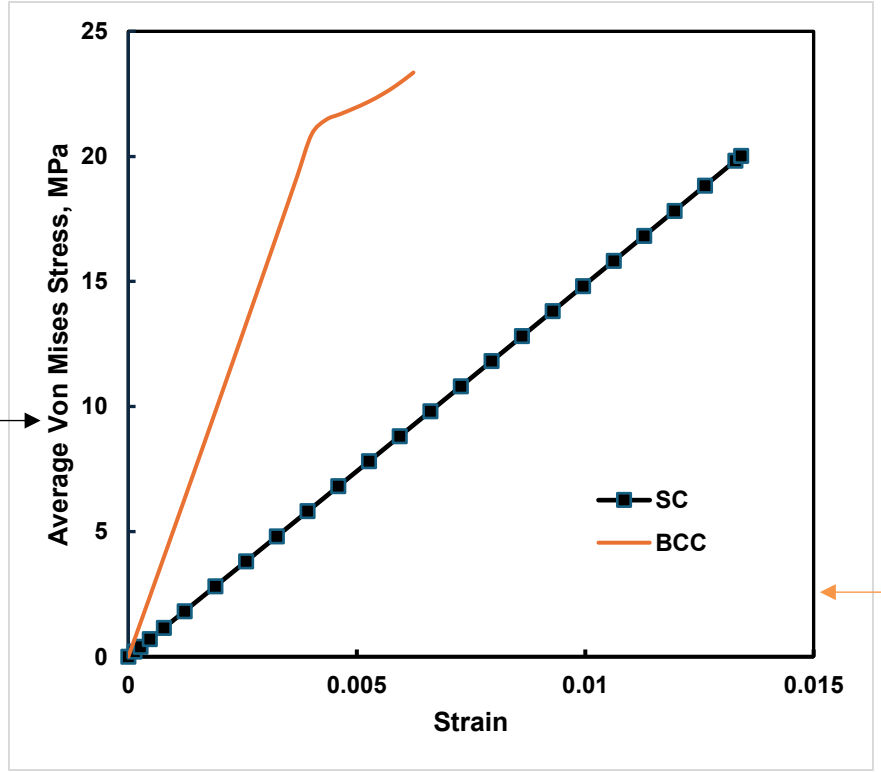

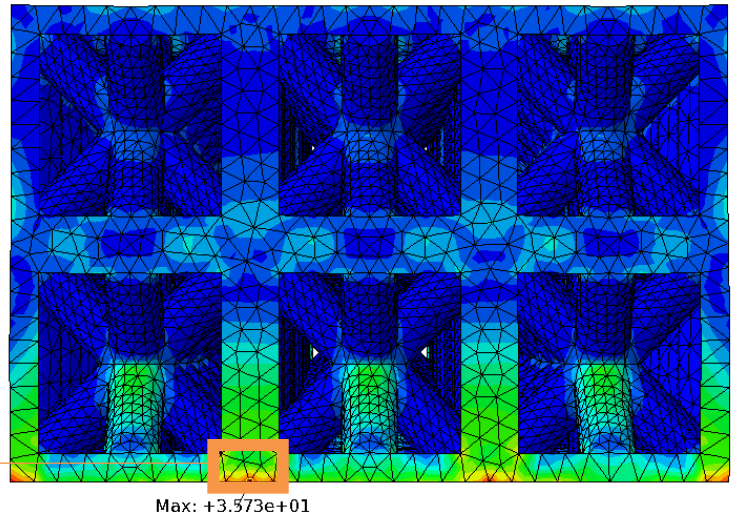

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